

VisionClaw Agent Platform

AI Buddy LLC — Illinois
Comprehensive Features (March 2026)

VISIONCLAW AGENT PLATFORM

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Comprehensive Platform Features (March 2026)

Platform Stats:

- 473 project files (~452,000 lines of TypeScript/React)
- 66 PostgreSQL database tables (Drizzle ORM + pgvector)
- 96 AI tools (built-in + custom tool creation)
- 14 AI personas operating as a complete corporate team
- 36+ AI models across 8+ providers
- 23 active skills injected into agent system prompts
- 40 governance rules across 7 categories
- 11 research programs (5 nightly + 6 business)
- 7 research schedules
- 13 heartbeat tasks
- 38+ frontend pages

AUTHENTICATION & MULTI-TENANCY

- Replit OAuth sign-in (primary)
- Email + password with verification flow
- Admin PIN (HMAC-SHA256 with salt)
- DB-backed sessions (17+ active)
- Strict multi-tenant data isolation
- Per-tenant memory backup
- Per-tenant file storage (Replit Object Storage)
- Timing-safe cryptography
- Password policy enforcement
- Soft account deletion

14 AI PERSONAS — THE AI CORPORATION

1. VisionClaw — Platform Core Engine
2. Felix — CEO Orchestrator & Primary Presenter
3. Forge — Code Generation & Engineering
4. Teagan — Testing & QA Specialist
5. Blueprint — Architecture & Design
6. Chief of Staff — Operations & Stability
7. Scribe — Documentation & Communication
8. Proof — Output Quality Reviewer (Auto-QA)
9. Radar — Research & Web Intelligence
10. Neptune — Data Analysis & Visualization
11. Apollo — Creative & Content Generation
12. Atlas — Strategic Planning & Advisory
13. Cassandra — Finance & Market Intelligence
14. Luna — Customer Relations & Support

Each persona has:

- Specialized system prompts with platform briefing

- Dedicated tools and skills
- Configurable reasoning modes (auto-thinking)
- Cost tier preferences (free !' cheap !' paid)
- Cross-persona knowledge via vector search
- Tools documentation (auto-synced via sync_personas)

96 AI TOOLS

Communication: send_email, send_whatsapp, send_discord, send_telegram
 Research: web_search, deep_research, trend_research, competitive_analysis
 Documents: generate_pdf, create_slides, export_persona
 Code: code_execute, code_review, generate_code
 Virtual Browser: browse_url, screenshot, fill_form, click_element
 Web Scraping: firecrawl_scrape, firecrawl_crawl, firecrawl_map, jina_read
 Agentic: delegate_task, orchestrate, strategic_interview
 Google Workspace: gdrive_upload, gdrive_list, gdrive_search
 Charts & Dashboards: generate_chart, generate_dashboard
 Media: generate_social_image, produce_video, generate_audio
 Finance: market_news, stock_data, stock_search, market_overview
 Diagrams: render_diagram (Mermaid !' PNG)
 Voice: vibevoice_transcribe (VibeVoice ASR, 60-min, 50+ languages)
 System: sync_personas, health_check, manage_skills
 + Custom tool creation with auto-persona-sync

SMART MODEL AUTO-ROUTING

OAuth-First / Subscription-First (BYOS):

- Google Gemini OAuth (PKCE + generative-language scope)
- OpenAI ChatGPT Plus (code-paste + STS exchange)
- Claude Runner bridge (\$0 per-token via Max plan)

Model Tiers:

- Fast: Gemini 3 Flash !' Gemini 2.5 Flash !' GPT-4.1 Mini
- Balanced: Gemini 3 Flash !' GPT-4.1 !' Sonnet 4
- Powerful: Gemini 3.1 Pro !' GPT-4.1 !' Opus 4.6 !' Sonnet 4.6
- Reasoning: Gemini 3.1 Pro !' o4-mini !' Opus 4.6

Features:

- Task Complexity Classifier
- Multimodal-Aware Routing
- Auto-Thinking Mode
- Persona Cost Tier Integration
- Model Capabilities Registry
- [auto-route] logging with 60s dedup
- markSubscriptionFailed with tiered TTLs
- All 36 models tagged: free/cheap/paid

AUTONOMOUS OPERATIONS

Heartbeat Engine:

- 13 recurring tasks running on configurable intervals
- Auto-backup to Google Drive
- Corporation report export (PDF)
- GitHub auto-push with secret scanning
- Research schedule execution
- Health monitoring (6 checks)

Chief of Staff Stability Watchdog:

- Auto-remediation of stuck tasks
- Flaky task detection
- Memory pressure monitoring
- Stale data cleanup
- Pool health monitoring

Felix Auto-Orchestration:

- Multi-step task decomposition into DAG
- Specialist persona assignment
- Parallel and sequential execution
- Child conversation delegation

Human-in-the-Loop (HITL):

- Confirmation gate for sensitive operations
- Felix approval gate for high-impact decisions

NIGHTLY AUTORESEARCH SYSTEM

Inspired by Karpathy's Autoresearch:

- 11 autonomous research programs
- 5 nightly platform programs (2 AM daily)
- 6 AI Buddy business programs (Sunday midnight CT)
- Keep/discard loop with scoring ("e6 = keep)
- Executive summaries per session

Auto-Deposit Intelligence Loop:

- Findings auto-route to correct project folder
- Auto-inject into agent_knowledge with vector embeddings
- High-score findings ("e8) generate code proposals
- Programs 2-7 !' AI Buddy Health project
- Programs 8-12 !' VisionClaw Agent Platform project

Adaptive Backpressure:

- Level 1 (1-2 timeouts): 2x slower
- Level 2 (3-5 timeouts): 3x slower + reschedule
- Level 3 (6+): pause ALL research 5 min
- Max 2 concurrent sessions enforced

GOVERNANCE & SECURITY

40-Rule Process Governor:

- 7 categories: data, finance, comms, system, research, legal, operations
- 25 condition evaluators
- Emergency kill switch
- NIST, OWASP, Singapore IMDA compliance

Agency Expansion Framework (6 Tiers):

- 9 evaluator types
- Trust Score Engine (9 categories)
- Proactive Initiative Engine (PAB budgets)
- 12 Express Lanes
- 8 Environmental Awareness scan types
- 4 Collective Intelligence protocols

Security:

- Helmet CSP headers
- Provider Key Proxy (no client exposure)
- DB-persisted password reset tokens
- IronClaw-inspired SafetyLayer
- Per-tool sliding-window rate limiter
- Execution Supervisor: circuit breaker, output validation, hallucination detection

INTELLIGENCE & MEMORY

Memory Architecture:

- Hierarchical Memory Graph
- Three-Tier Semantic Memory
- BM25/Vector/Hybrid Document Search
- Zero-Loss Compaction with safety gate
- pgvector for native PostgreSQL vector similarity
- Embedding cache (30s TTL, max 50 entries)

Instinct Learning System:

- Auto-extracts reusable patterns from successful tasks
- Confidence scoring and graduation (3+ observations, >70%)
- Graduated instincts injected into system prompts
- Agents get progressively better at recurring tasks

Cross-Persona Knowledge:

- Vector search pulls findings from ANY persona
- 2000-char knowledge budget per prompt
- Persona-specific entries ranked first

Project Brain:

- Auto-maintained .md knowledge file per project
- Injected into project conversations
- Auto-transcript system with asset capture

PRESENTATION & VISUALIZATION

- Mermaid Diagram Rendering (flowcharts, architecture maps, etc.)
- Live Delegation Event Feed (real-time SSE)
- Voice Narration for agent activity (browser TTS, free)
- Chart Generation (bar, line, pie, area via Recharts)
- Interactive Dashboards (HTML/CSS/JS in live canvas)
- AI Image Generation (auto-upload to Drive)
- Video Production (TTS narration + MP4)
- Audio Generation (OpenAI TTS primary)
- PowerPoint Slide Generation (2slides API)
- VibeVoice ASR (60-min speech-to-text, 50+ languages)

DELEGATION & MULTI-AGENT

Multi-Layer Delegation:

- Configurable depth up to 5 levels
- Tool restrictions scale by depth
- Depth 2+ blocks spawn/subagents
- Depth 3+ blocks orchestrate
- Depth 5 blocks all delegation
- Trust/expansion context at depth "d1"

Live Activity Feed:

- Real-time SSE streaming
- Tenant-isolated event routing
- 5-minute TTL auto-cleanup
- Color-coded agent icons
- Depth indicators and timestamps
- Voice narration in plain English

Auto-QA Pipeline (De-Sloppify):

- Proof (persona 8) auto-reviews delegation output
- Scores: completeness, accuracy, clarity, professionalism
- Non-blocking fire-and-forget execution
- Color-coded verdict badges in UI

Per-Task Cost Tracking:

- Real-time USD, token counts, duration
- Model breakdown per delegation step
- Cost badges in live delegation feed

COMMUNICATION & INTEGRATIONS

Channels:

- AgentMail (visionclaw@agentmail.to)
- WhatsApp Web (Baileys library)
- Discord Bot
- Telegram Bot
- Email with tenant inboxes

Payments:

- Stripe Connect (subscriptions, BYOK tiers)
- Coinbase CDP SDK + Commerce API
- Crypto payment support
- Usage metering

Storage & Files:

- Replit Object Storage (per-tenant)
- Google Drive integration (upload, share, folders)
- PDF toolkit
- File Manager UI

External Services:

- Firecrawl (web scraping, crawling, site mapping)
- Jina AI (Reader)
- ip-api.com (geolocation)
- Open-Meteo (weather)
- mermaid.ink (diagram rendering)

SELF-IMPROVEMENT ENGINES

Autonomous Self-Tuning:

- Runs every 24 hours
- 7-day performance metrics collection
- Parameter adjustment within safe bounds
- Bootstrap mode for new tenants
- Dynamic express lane caps based on trust

Instinct Learning:

- Pattern extraction from multi-tool successes
- Graduation after 3+ observations at >70% confidence
- Permanent knowledge injection

Persona Sync (sync_personas — Tool #96):

- Auto-syncs all 14 persona tools_doc and agents_doc
- Triggers on skill toggle and custom tool creation
- Mutex prevents concurrent sync races
- Admin-only execution scope
- Per-persona error handling

Code Proposals:

- High-score research findings ("e8) auto-generate proposals
- Stored in code_proposals table
- API: GET/PATCH /api/research/code-proposals

TECH STACK

Frontend:

- React 18, Vite, TypeScript
- shadcn/ui, TailwindCSS
- Wouter (routing), TanStack Query v5
- Framer Motion (animations)
- Recharts (charts)

Backend:

- Express.js, TypeScript
- Drizzle ORM, Zod validation
- Helmet (security headers)
- Server-Sent Events (SSE)

Database:

- PostgreSQL with pgvector
- 66 tables
- Drizzle ORM with Zod schemas

AI Providers:

- OpenAI (OAuth + direct + Replit integration)
- Anthropic (Claude Runner bridge + direct + integration)
- Google Gemini (OAuth + integration)
- xAI (Grok)
- OpenRouter (DeepSeek, MiniMax, Qwen, Llama, etc.)
- ElevenLabs (TTS/STT)

Infrastructure:

- Replit hosting
- Replit Object Storage
- GitHub (version control)
- Google Drive (file storage)

FRONTEND PAGES (38+)

- Command Center Dashboard
- Chat Interface with delegation live feed
- Agent Management & Configuration
- Persona Settings & Skills
- Research Programs & Sessions
- Heartbeat Monitor
- Governance Rules
- Trust Scores
- Usage Dashboard
- File Manager
- Project Manager with brain system
- API Keys & Provider Settings
- Billing & Subscriptions
- Admin Panel
- User Settings
- Onboarding (3-step)
- Legal pages (Terms, Privacy, Cookie Policy)
- Error boundary pages

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